

Experiment 40

Aim

To draw rectangular shapes around objects and extract the detected objects from an image using OpenCV.

Procedure

1. Import OpenCV.
2. Read the input image.
3. Convert the image to grayscale.
4. Apply thresholding to separate objects from the background.
5. Find the contours of the objects.
6. Find the bounding rectangle for each contour.
7. Draw rectangles around the detected objects.
8. Extract the objects using the coordinates of the bounding rectangles.
9. Display and save the result.

Code:

```
import cv2

img = cv2.imread("image.jpg")

if img is None:

    print("Error: Image not found!")

    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

_, thresh = cv2.threshold(gray, 120, 255, cv2.THRESH_BINARY)

contours, _ = cv2.findContours(
```

```
    thresh,

    cv2.RETR_EXTERNAL,

    cv2.CHAIN_APPROX_SIMPLE

)

count = 0

for contour in contours:

    if cv2.contourArea(contour) < 1000:

        continue

    x, y, w, h = cv2.boundingRect(contour)

    cv2.rectangle(

        img,

        (x, y),

        (x + w, y + h),

        (0, 255, 0),

        2

    )

    object_img = img[y:y+h, x:x+w]

    cv2.imwrite("object_" + str(count) + ".jpg", object_img)

    count += 1

cv2.imshow("Objects with Rectangles", img)
```

```
cv2.imwrite("rectangular_objects.jpg", img)
```

```
cv2.waitKey(0)
```

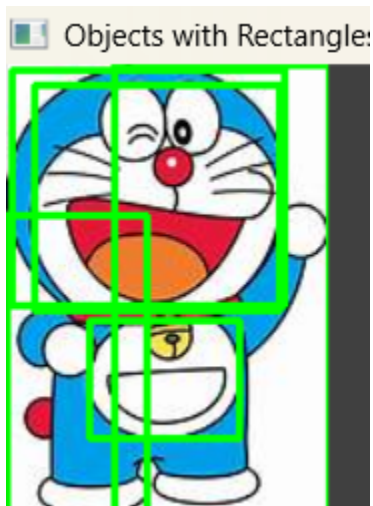
```
cv2.destroyAllWindows()
```

```
print("Number of objects extracted:", count)
```

Sample Input:



Output:



Result

The objects in the given image were successfully detected using OpenCV. Rectangular bounding boxes were drawn around the detected objects, and each detected object was extracted